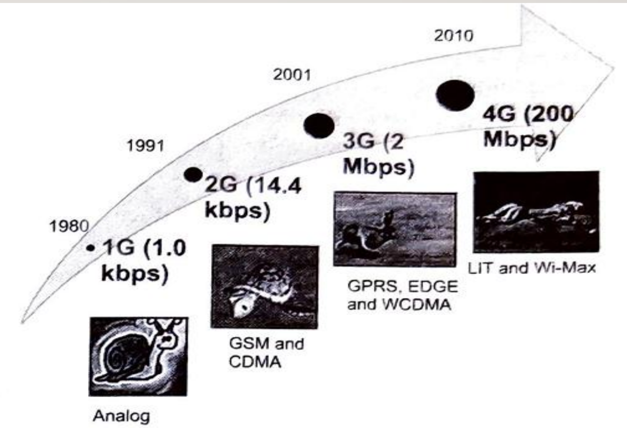
MOBILE ANALYTICS

INTRODUCTION TO MOBILE ANALYTICS

- Mobile Analytics is the process of collecting, analyzing, and interpreting data from mobile apps and devices to understand user behavior, app performance, and marketing effectiveness.



Source: *3GPP Alliance, UMTS forums, Informa telecoms, Motorola, ZTI.*

Figure 29.1: Evolution of Mobile Technologies

TYPES OF APPLICATION FOR MOBILE ANALYTICS

1. *MOBILE WEB ANALYTICS*
2. *MOBILE APPLICATION ANALYTICS*

MOBILE WEB ANALYTICS

Refers to the use of mobile phones or tablets to view online content.

- Mobile Web uses a lightweight browser and often has sites like m.example.com.
- Website design depends on screen size—large screen designs may not suit mobiles.
- Some organizations build sites specifically for tablets.
- Responsive design ensures the website adapts to different screen sizes.

MOBILE APPLICATION ANALYTICS

- Refers to analytics of mobile apps that run on smartphones or tablets.
- Apps are distributed through platforms like Google Play and Apple App Store.
- Apps can be free or paid depending on their functionality.
- Used when organizations need high engagement and interactivity.
- Mobile websites are better for business purposes; apps for user engagement.

DIFFERENCES BETWEEN MOBILE APP ANALYTICS AND MOBILE WEB ANALYTICS

Factors	Mobile App Analytics	Mobile Web Analytics
Screen and Page	Does not have pages; the user interacts with various **screens**.	Has pages like a normal **website**, and users interact with multiple pages.
Use of Built-in Features	Can access built-in mobile features such as **GPS, gyroscope, accelerometer, and storage**.	Does **not use built-in features** like GPS or accelerometer.
Session Time	Has **shorter session timeouts** (around **30 seconds**).	Has **longer session timeouts** (usually ends after about **30 minutes** of inactivity).
Online / Offline Functionality	May **work offline** depending on how the app is developed; may not require a network connection.	Requires an **Internet connection**; can only run online.
Updates	App owners provide **frequent updates** and new versions.	Updates are **less frequent** compared to apps.
Platform Availability	Distributed through **app stores** (Google Play, Apple App Store, etc.).	Accessed via **web browsers** (e.g., Chrome, Safari).
Engagement Type	Provides **interactive and personalized experiences**.	Focuses more on **informational or business-related purposes**.
Development Cost	Generally **higher development and maintenance cost**.	**Lower cost** and easier to maintain across devices.

CLASSES OF MOBILE ANALYTICS TOOLS

- 1. Internal Mobile Analytics Tools – Installed and maintained by IT departments (e.g., Localytics, WebTrends).
- 2. External Mobile Analytics Tools – Provided by third-party vendors for data collection and analysis (e.g., comScore, Groundtruth).

TOP MOBILE ANALYTICS TOOLS

- **1. Localytics** – Marketing and analytics platform supporting push messaging and business analytics.
- **2. Appsee** – Provides heatmaps, conversion analysis, and user interaction insights.
- **3. Google Analytics** – Free service for app and web analytics.
- **4. Mixpanel** – Business analytics service focusing on user engagement and behavior tracking.

CATEGORIES OF MOBILE ANALYTICS TOOLS

- 1. Location-based Tracking Tools
- 2. Real-time Analytics Tools
- 3. User Behavior Tracking Tools

LOCATION-BASED TRACKING TOOLS

- These tools store and analyze location information from mobile devices.
- Examples:
- **Geoloqi** – Platform for location-based services and private location sharing.
- **Placed** – Provides ratings service to analyze offline consumer behavior.

REAL-TIME ANALYTICS TOOLS

- Real-time analytics tools analyze and report user data instantly.
- Example:
- **Geckoboard** – Real-time dashboard that collects, displays, and reports key business data.

USER BEHAVIOR TRACKING TOOLS

- These tools track user interactions within apps such as:
- Screens viewed, session length, technical errors, clicks on ads.
- Example:
- **TestFlight** – Used by developers to test apps, collect feedback, and track user behavior.

PERFORMING MOBILE ANALYTICS

Steps:

1. Define objectives and identify target audience.
2. Collect and prepare data.
3. Apply data mining techniques (e.g., text mining).
4. Analyze, evaluate, and interpret results.

DATA COLLECTION IN MOBILE ANALYTICS

1. **Data Collection through Mobile Device** – Real-time data collection and tracking (e.g., Numbers, StatViz).

2. **Data Collection on Server** – Data sent to servers for analysis (e.g., DataWinners).

CHALLENGES IN MOBILE ANALYTICS

1. Data Privacy Concerns

- Managing sensitive user data under GDPR and other regulations.

2. Data Fragmentation

- Handling multiple devices, operating systems, and versions.

3. Real-Time Analysis Complexity

- Processing large amounts of live user data.

4. Attribution Tracking

- Identifying the exact source of app installs or conversions.

CONCLUSION

- Mobile Analytics provides powerful insights into user behavior and app performance. It helps businesses enhance customer experience, optimize marketing strategies, and improve overall mobile application efficiency.